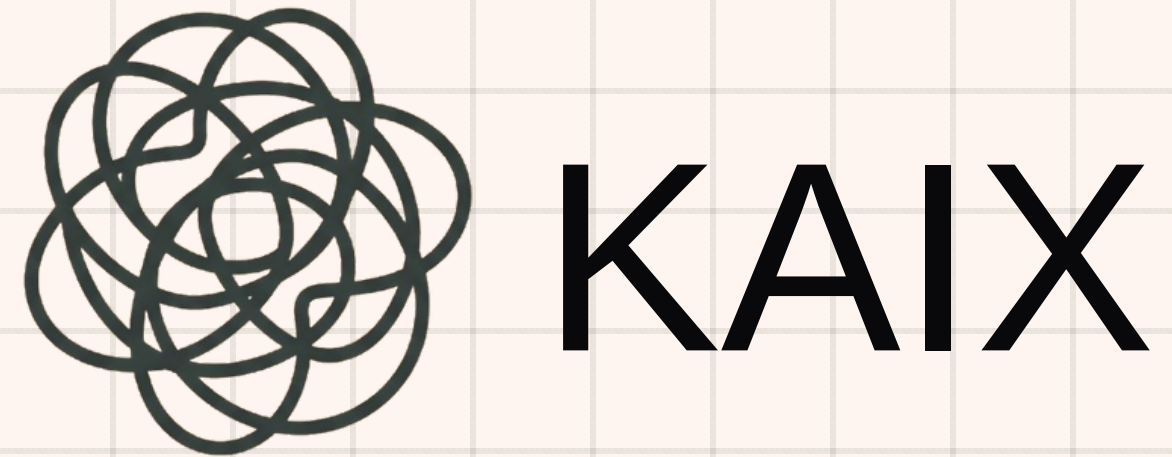




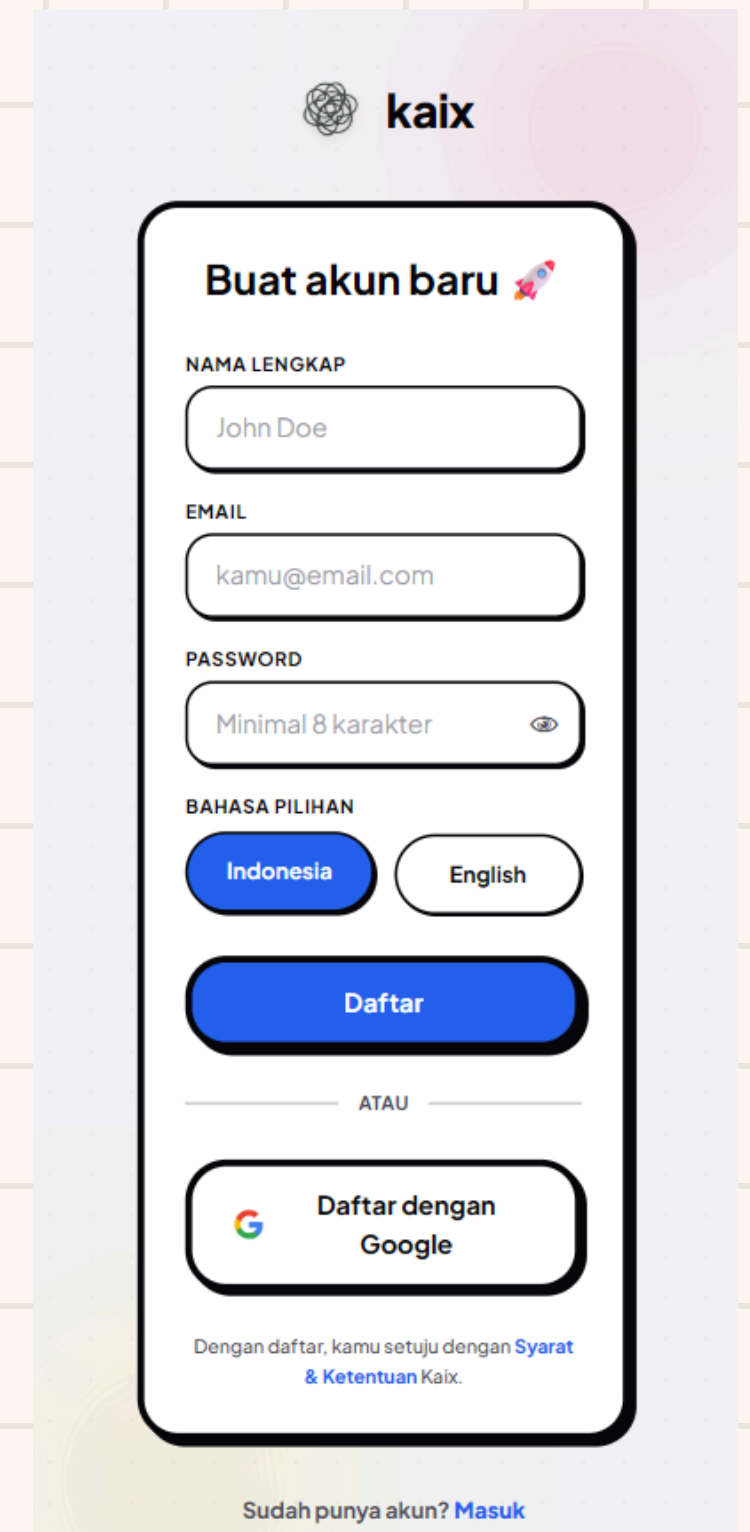
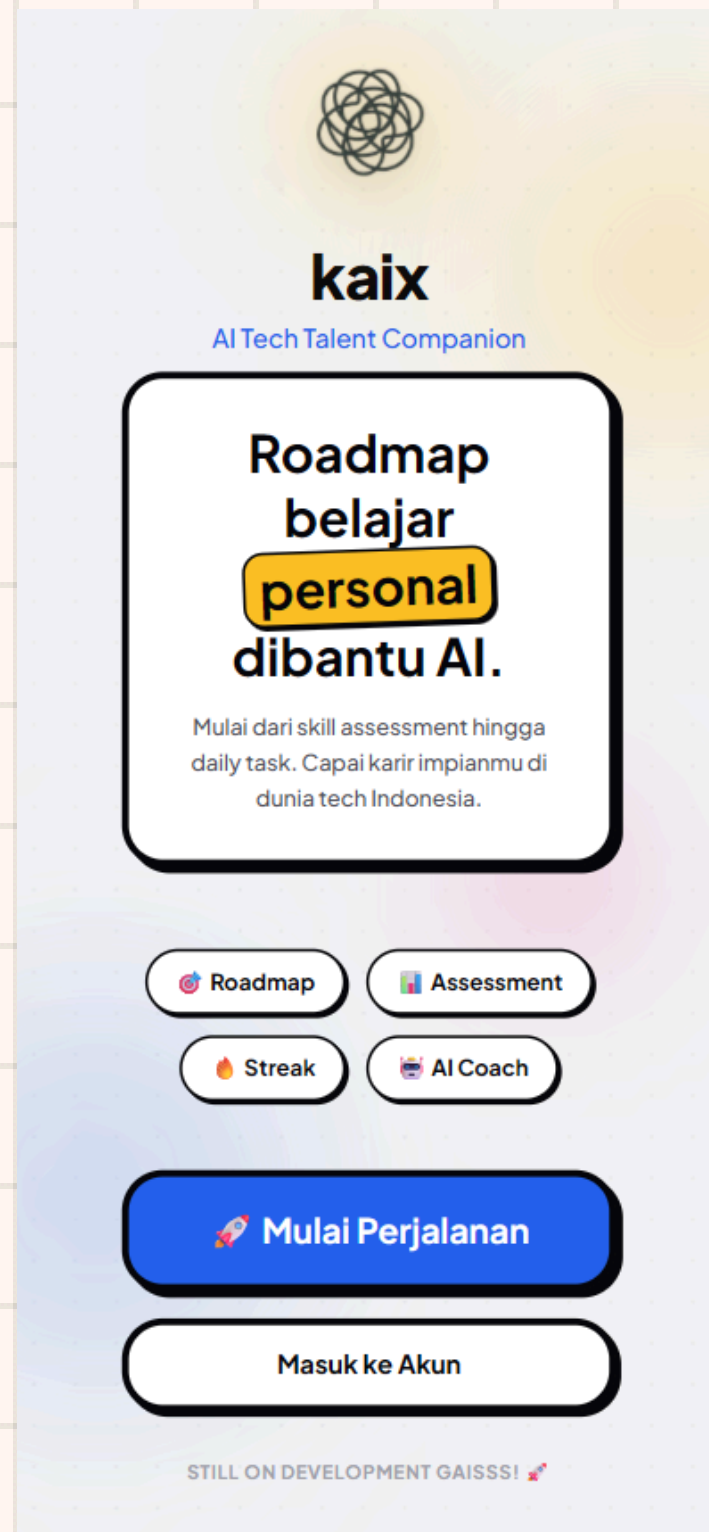
AI TECH TALENT COMPANION 

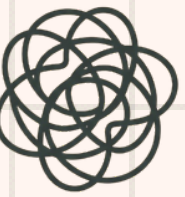
Roadmap belajar **personal**, dibantu AI.
Mulai dari skill assessment hingga daily task.
Capai karir impianmu di dunia tech Indonesia.

 MULAI PERJALANAN



github.com/Haniff-Toha/kaix-ai-tech-talent-dev

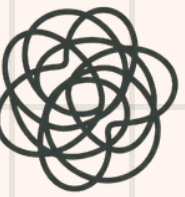




MASALAH: BELAJAR OVERWHELMING

Masalah	Dampak
Terlalu banyak resource	Paralysis of choice — bingung mulai dari mana
Tidak ada roadmap personal	Belajar acak tanpa arah, skill gap tak teridentifikasi
Tidak ada accountability	Mudah putus di tengah jalan, streak belajar tidak terjaga
Konten tidak relevan	Resource Bahasa Inggris, tidak sesuai konteks Indonesia

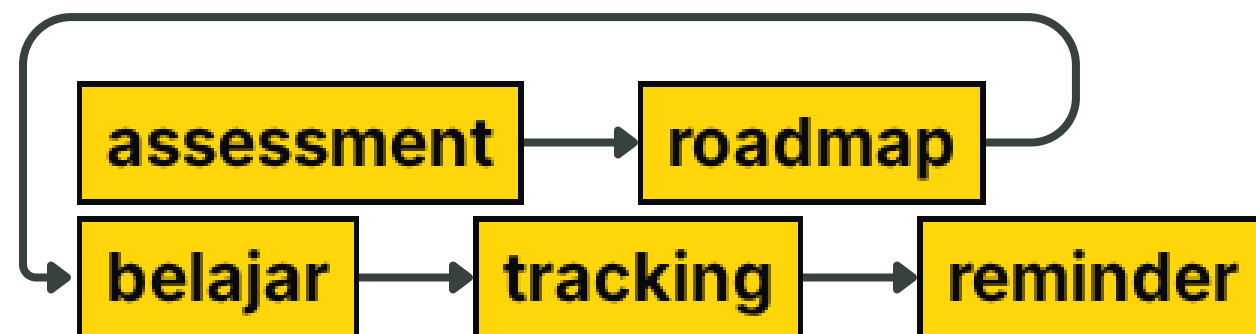
85% self-learner di Indonesia berhenti belajar dalam 3 bulan pertama karena tidak punya panduan personal dan sistem accountability.



SOLUSI: AI LEARNING COMPANION

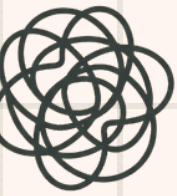
PLATFORM TERPADU

Satu platform untuk



Semua di-power oleh AI yang memahami konteks karirmu.

- | | |
|----------------------------------|-----------------------|
| 🎯 Profile & Skill Assessment | ← Profile Agent |
| 📖 Personalized Roadmap | ← Roadmap Agent + RAG |
| ⚡ Focus Timer + Activity Logging | ← Logging Classifier |
| 🔔 Smart Reminders & Nudges | ← Nudge Agent |
| 📊 Progress Tracking & Streaks | ← Gamification Layer |



FITUR 1: ONBOARDING &

AI memahami posisimu dan target karirmu

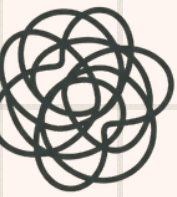
1 User mengisi form: role, target role, skills, pengalaman.

2 Profile Agent menganalisis gap skill saat ini vs target.

3 Menghasilkan summary, strengths, skill gaps, & timeline.

4 Profil disimpan & di-embed ke User RAG untuk konteks masa depan.

Tech: (gpt-oss-120b) + Pydantic + pgvector RAG



FITUR 2: AI-GENERATED ROADMAP

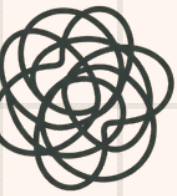
CARA KERJA

- SQL Query → Ambil profil user & target role
- Knowledge RAG → Cari konten yang relevan
- LLM Generation → Generate JSON terstruktur
- Storage → phases → milestones → tasks

Pola: SQL Agent + RAG → LLM Generation

OUTPUT ROADMAP

```
{
  "total_phases": 3,
  "estimated_months": 6,
  "phases": [
    {
      "phase_number": 1,
      "phase_title": "Fundamental Deep Learning",
      "milestones": [
        {
          "title": "Statistik & Matematika",
          "skills": ["Statistics", "Linear Algebra"],
          "target_hours": 30,
          "daily_tasks": [...]
        }
      ]
    }
  ]
}
```

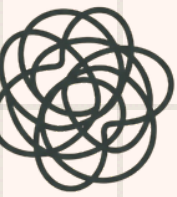



FITUR 3: REKOMENDASI KURSUS

Sumber	Tipe	Jumlah Data
Dicoding	Learning paths Indonesia	12 paths, 50+ kelas
YouTube	Video tutorial (ID & EN)	100+ video per skill
rubythalib.ai	Curated AI/ML courses	7 kursus spesifik
Books	Buku referensi	5 buku esensial

PIPELINE

Scraper → Normalizer → RAG Chunk Generator → Embedding → pgvector



AI DEEP DIVE: RAG PIPELINE

DUAL-RAG ARCHITECTURE

Knowledge RAG

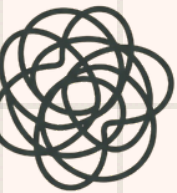
Scraped Courses, Books, Career Content
career_track filter + vector search
→ pgvector cosine similarity

User RAG

User's profile, activity history,
classified skills, progress notes
user_id scoped + vector search

KEY DESIGN DECISIONS

- Embedding Model: Gemini embedding-2-preview (MRL-trained → safe truncation)
- Vector DB: Supabase PostgreSQL + pgvector extension
- Similarity: Cosine distance ($\leq$)
- Task Type Hints: RETRIEVAL_DOCUMENT & RETRIEVAL_QUERY
- Dimension: 768 (Supabase pgvector limit is 2000)

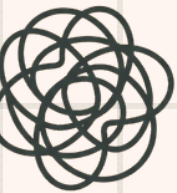


AI DEEP DIVE: AGENT SYSTEM



4 SPECIALIZED AGENTS

Agent	Model	Output
Profile	Groq 120b	Pydantic Analysis
Roadmap	Groq 120b + RAG	JSON Roadmap
Classifier	llama-3.3-70b	Activity Map
Nudge	llama-3.3-70b	Nudge Message



AI DEEP DIVE: LLM SERVICE

AUTO FAILOVER

```
class LLMService:
    primary: BaseChatModel # gpt-oss-120b
    fallback: BaseChatModel # llama-3.3-70b

    # Primary → Fallback automatic routing
    async ainvoke()

    # JSON schema enforcement
    async ainvoke_with_structured_output()

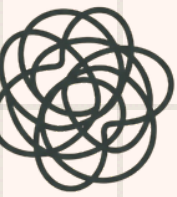
    # Streaming with fallback
    async astream()
```

Reliability:

Primary fails → fallback auto-triggers → zero
downtime.

MODEL TIERS

Tier	Model	Use Case
Reasoning	gpt-oss-120b	Profile, Roadmap
Fast	llama-3.3-70b	Classifier, Nudge
Vision	llama-4-scout-17b	Verification Agent
Embedding	gemini-2-preview	Vectors (768 dims)



FITUR 4: FOCUS TIMER & LOGGING

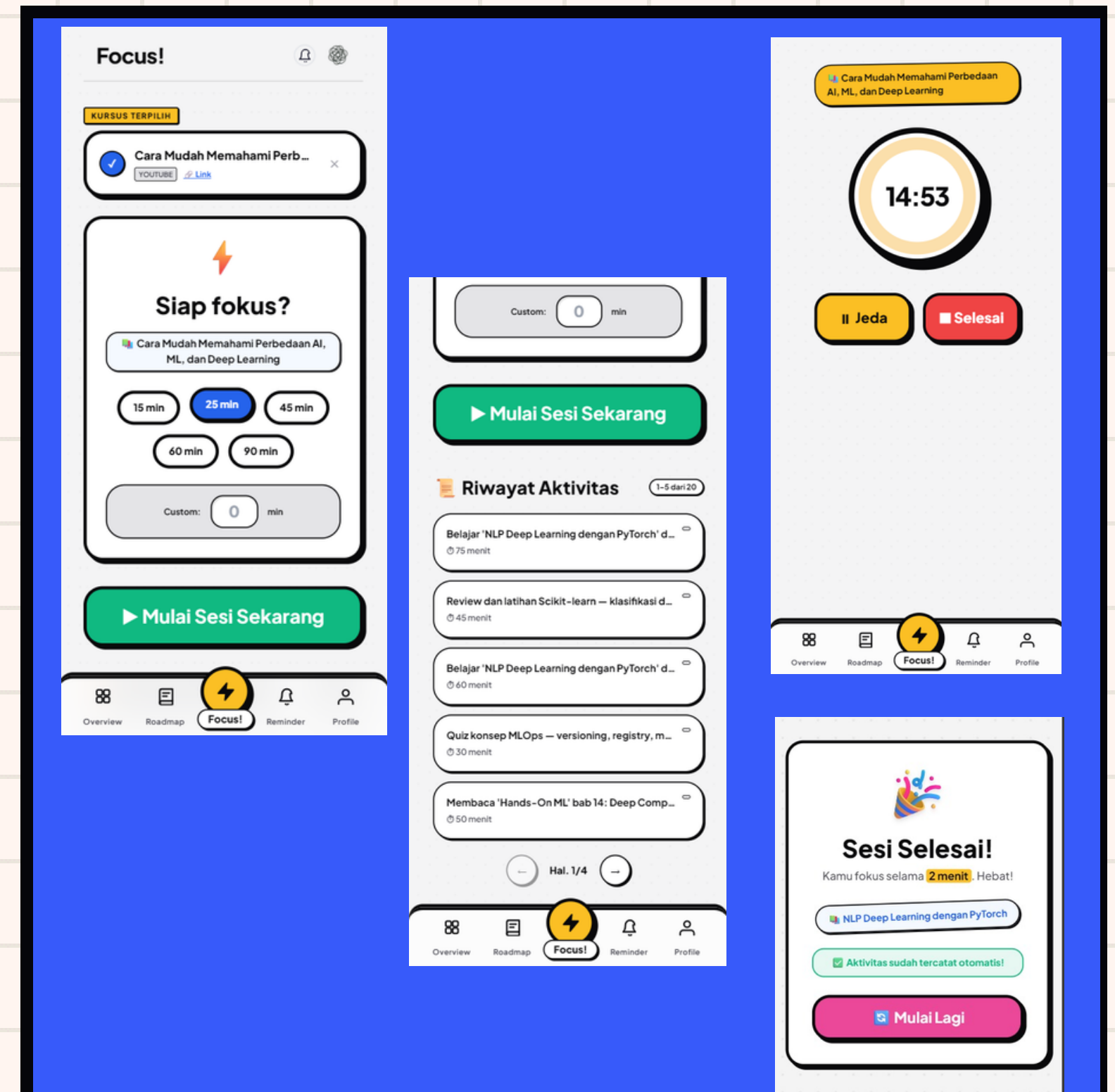
🕒 User pilih kursus / fokus bebas & set durasi.

▶ Timer berjalan → pause/resume → auto-log.

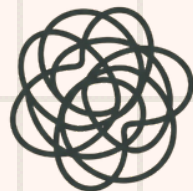
🤖 Logging Classifier mengklasifikasi aktivitas ke milestone roadmap.

Input: "Belajar NLP dengan PyTorch"

```
Output: {
  classified_skill: "NLP",
  mapped_milestone_id: "m003",
  skill_level_signal: "learning"
}
```



FITUR 5: SMART REMINDERS



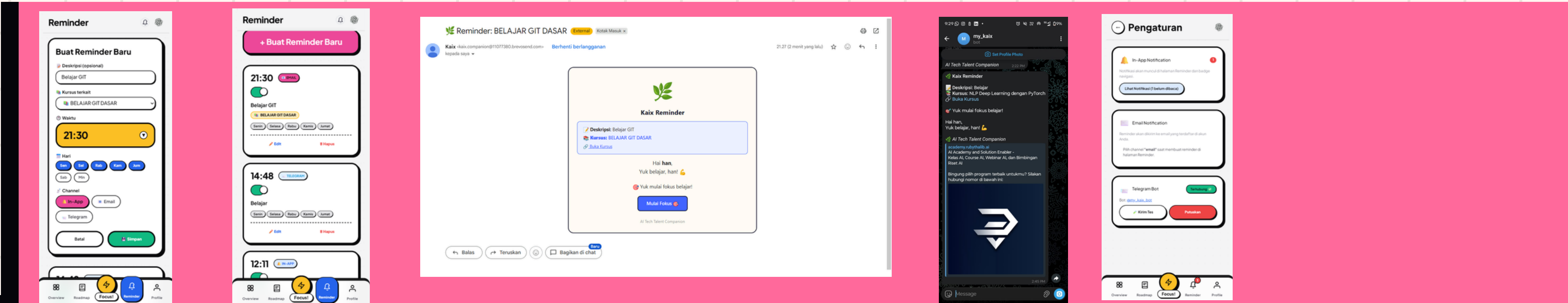
Channel	Status	Deskripsi
In-App	Live	Notifikasi di dalam aplikasi
Email	Live	Email dengan konten kursus
Telegram	Live	Bot integration via webhook

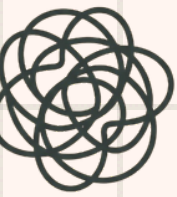
SCHEDULER FLOW

APScheduler (CronTrigger/min)
↓ Check due reminders
↓ Nudge Agent (Generate Msg)
↓ Dispatch to Channel

"Han, sudah waktunya review materi NLP hari ini. Yuk lanjutkan progress-mu!"

Course: NLP Deep Learning dengan PyTorch





FITUR 6: PROGRESS & GAMIFICATION

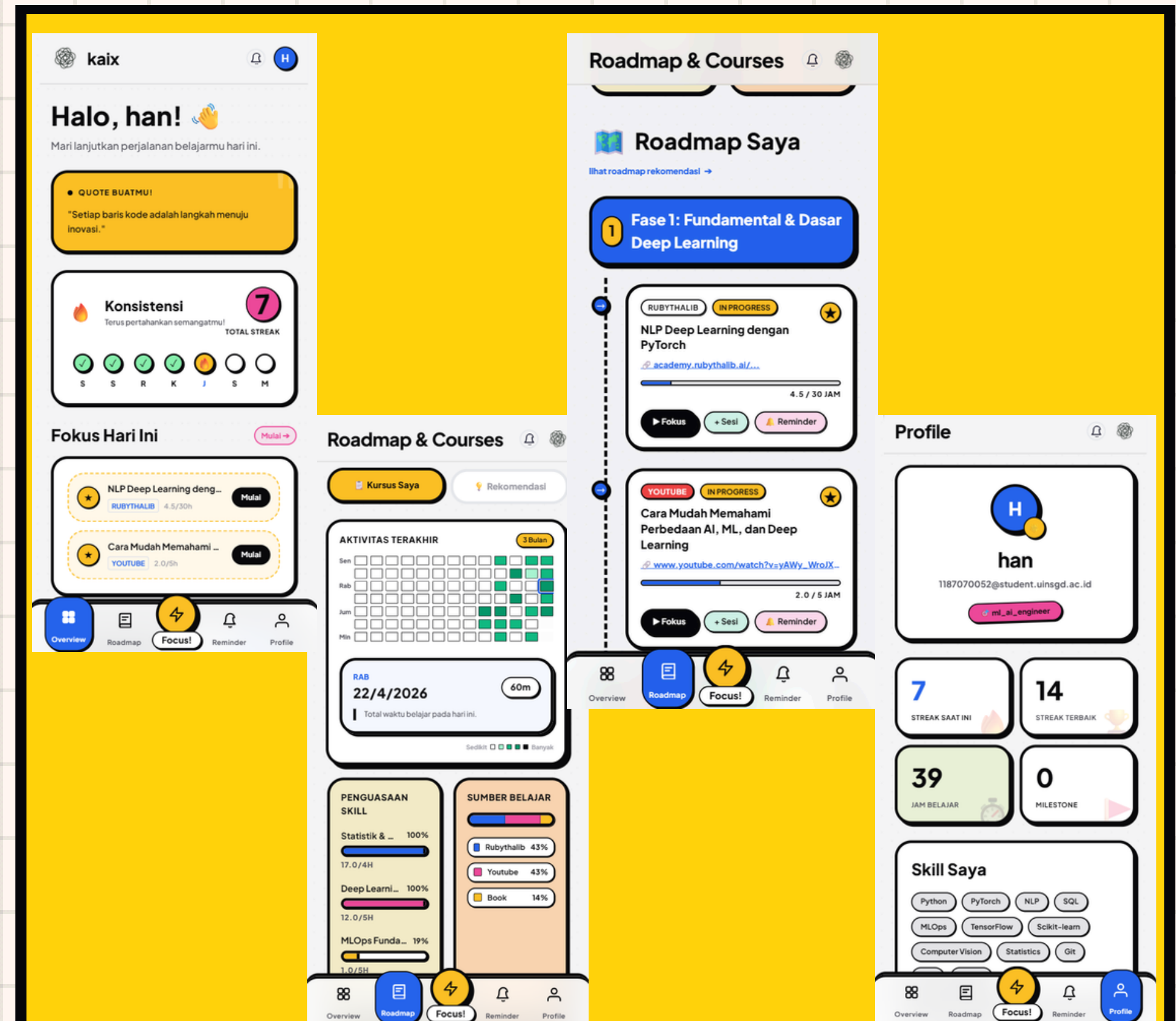
 Streak counter: hari berturut-turut belajar.

 Activity heatmap: visualisasi 3 bulan grid.

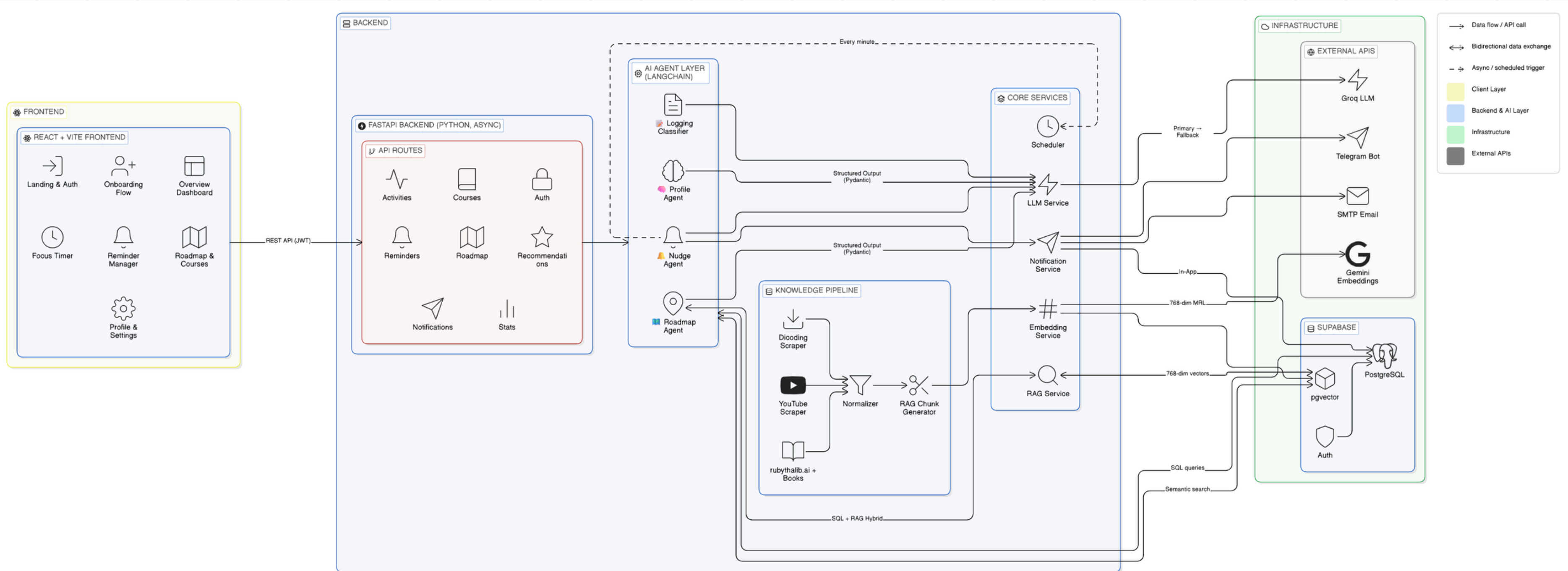
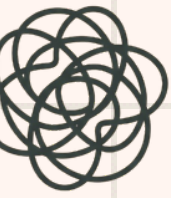
 Skill mastery: progress bar per milestone.

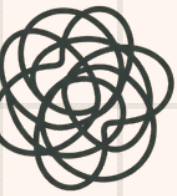
Mastery Calc:

```
milestone_progress += delta (max 0.2/act)
mastery % = (accum_hours / target_hours) * 100
```



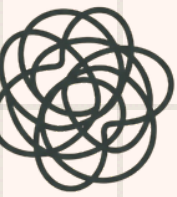
FULL STACK ARCHITECTURE





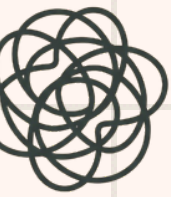
TECH STACK

Layer	Technology	Rationale
Frontend	React + Vite + Tailwind	Fast dev, mobile-first
UI Framework	Neo-brutalist design	Distinctive, memorable brand
Animation	Framer Motion	Smooth micro-interactions
Backend	FastAPI (Python)	High performance, async-native
ORM	SQLAlchemy 2.0	Type-safe, production-grade
Database	Supabase PostgreSQL	Managed, auth built-in
Vector DB	pgvector (768 dims)	No extra infra, co-located
LLM Provider	Groq	Ultra-fast inference, free tier
Embedding	Gemini 2 preview	MRL for dimension reduction
AI Framework	LangChain	Structured output, failover
Scheduler	APScheduler	In-process, reliable
Auth	Supabase Auth + JWT	Social login ready
Migrations	Alembic	Database versioning



AUTOMATED KNOWLEDGE PIPELINE





ALGORITHM SPOTLIGHT

Pola Hybrid: SQL Agent + RAG



Step 1: SQL Query

Ambil user profile, roadmap, activity history (Structured, exact, relational).



Step 2: RAG Search

Query knowledge base via vector similarity (Fuzzy, semantic, contextual).



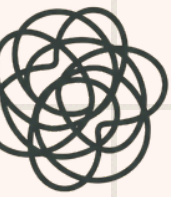
Step 3: Context Assembly

Gabungkan SQL data + RAG results sebagai context untuk AI agent.



Step 4: LLM Generation

Kirim ke LLM dengan structured output schema (Pydantic). Auto-failover.



EMBEDDING DEEP DIVE (MRL)

MATRYOSHKA STRATEGY

Problem: Supabase pgvector has a 2000-dimension limit.

Solution: Gemini embedding-2-preview uses MRL — truncated vectors maintain ~95% quality.

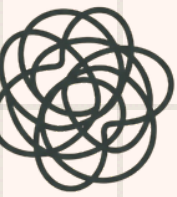
```
Full embedding: 3072 dims (too large)
MRL truncation: 768 dims (fits DB)
L2 norm:        preserved (cosine)
```

DUAL TASK TYPES

```
# Storage: optimize for doc retrieval
config = EmbedContentConfig(
    task_type="RETRIEVAL_DOCUMENT",
    output_dimensionality=768
)

# Query: optimize for query matching
config = EmbedContentConfig(
    task_type="RETRIEVAL_QUERY",
    output_dimensionality=768
)
```

Task-type hints improve retrieval quality by 5-15% vs generic embeddings.



STATUS: APA YANG SUDAH JALAN

Fitur	Status	Detail
Auth (Google + Email)	Li Live	Supabase Auth
Onboarding + Profile Agent	Li Live	Gap analysis, structured output
Roadmap Generation	Li Live	3-phase JSON roadmap
Course Scraping	Li Live	Dicoding, YouTube, rubythalib
Recommendation Engine	Li Live	Career-track filtered, SQL-based
Focus Timer	Li Live	Course-linked or free, auto-log
Activity Classifier	Li Live	Auto-classify to milestones
Streak System	Li Live	Daily streak tracking
Reminders	Li Live	In-App, Email, Telegram
Heatmap + Skill Mastery	Li Live	3-month heatmap, progress bars
Dual RAG	Li Live	pgvector, 768-dim embeddings
LLM Failover Service	Li Live	Primary → Fallback automatic

WHAT'S NEXT

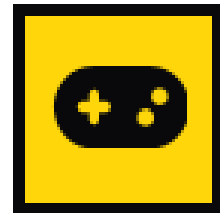


PHASE 2: INTELLIGENCE

Verification Agent: Vision LLM
memverifikasi screenshot.

Chat Assistant: Conversational
AI coach.

Weekly Recap: Rangkuman
otomatis.



PHASE 3: SOCIAL

Community: Leaderboard, study
group, peer accountability.

Achievements: Badge, level-up.

PWA: Offline support & Push.



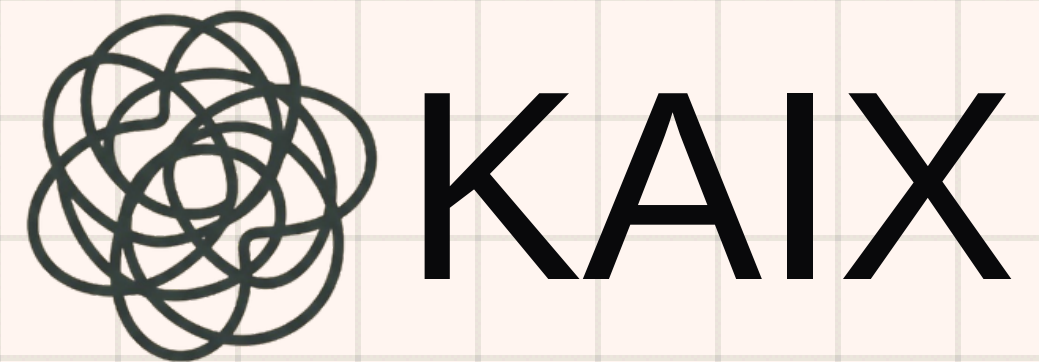
PHASE 4: SCALE

More Data: Coursera, Udemy,
bootcamps.

Hybrid Search: BM25 + Vector.

Analytics: Dashboard mentor.

Multi-Language: ID & EN.



AI TECH TALENT COMPANION

"Setiap baris kode adalah langkah menuju inovasi."

4 AI Agents

Dual RAG

Multi-Source Knowledge

LLM Fallback

768-dim Embeddings

React · FastAPI · Supabase · pgvector · Groq · Gemini · LangChain

Built with for Indonesia's tech talent